

# ABRIAN ABIR, M.S., E.I.T.

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## EDUCATION

### The University of Tennessee

Master of Science with Mechanical Engineering (*GPA: 4.00/4.00*)

Bachelor of Science in Mechanical Engineering; Minor in Computer Science (*GPA: 3.99/4.00*)

Knoxville TN

May 2026

May 2024

## LICENSES & CERTIFICATIONS

Engineer-in-Training (EIT), TN Department of Commerce and Insurance

Oct 15, 2025

Certified SOLIDWORKS Associate (CSWA), Dassault Systèmes

Nov 27, 2023

## EXPERIENCE

### Oak Ridge National Laboratory

Oak Ridge, TN

#### Applications Engineer for Advanced Manufacturing

Sep 2026 - Present

- Design, install, and calibrate advanced image-based and multi-sensor systems to ensure data integrity for machine learning applications.
- Develop custom data acquisition software and novel sensing techniques to upgrade and maintain optical/sensing infrastructure.
- Support cross-functional R&D projects for Manufacturing Demonstration Facility (MDF) aligned with core ORNL mission objectives.

#### Mechanical Systems Engineering Intern (SULI) – Fusion Research

May 2024 - Aug 2024

- Designed adaptive topology optimization framework to generate quantitative Figures of Merit (FoM), improving cooling performance.
- Automated multi-physics workflows by integrating HEAT and ANSYS Fluent using pyAnsys and PyGAD to reduce design iteration.
- Validated structural integrity of WEST tokamak probes and PFCs via coupled CFD/FEA under high heat-flux and transient EM loads.

### DENSO Manufacturing Tennessee, Inc.

Maryville, TN

#### Process Engineer I – Automation & Controls

Jun 2026 - Sep 2026

- Lead new product introduction, process planning, trials, and production capacity studies including line layouts and cost analysis.
- Manage new equipment specification, procurement, installation, and automation projects (including vision-guided robotics).
- Drive process optimization, root cause analysis, and IATF documentation for production improvements (PFMEA, PCS, Process Flows).

### University of Tennessee

Knoxville, TN

#### Graduate Researcher and Teaching Instructor

Aug 2024 - Jun 2026

- Engineered Python dissimilarity metrics for in-situ AFSD monitoring to enable automated defect detection from raw telemetry.
- Implemented thermal imaging for the live tracking of temperature during microwave power characterization for enclosed components.
- Instructed 30+ students in Mechanical Engineering Laboratory (ME 449), focused on data interpretation and technical documentation.

#### Undergraduate Researcher and Teaching Assistant

Nov 2020 - Aug 2024

- Designed & fabricated automated 3-point bend tester using Arduino-based custom software to characterize novel composite materials.
- Automated elastomeric 3D print buckling and deflection analysis using Python/OpenCV, reducing manual processing time.
- Mentored 60+ students in MATLAB and Engineering Fundamentals with faculty, managing project evaluations and weekly instruction.

#### Research Assistant Internship

Aug 2019 - May 2020

- Created Python visualization tools to standardize calculation of Young's modulus, UTS, and yield strength from raw MTS test data.
- Developed Python machine-vision applications (OpenCV) for automated inspection and visual tracking of thermoset 3D printing.

### Eastman Chemical Company

Kingsport, TN

#### Project & Controls Lead – Capstone Project

Aug 2023 - May 2024

- Led full-lifecycle development of film separator prototype, managing cross-functional team coordination & system-level requirements.
- Developed C++ closed-loop control software for dynamic motor operation, utilizing real-time sensor feedback to maintain film tension.
- Authored technical data package, from assembly drawings to specifications, to support the transition from prototype to production.

### GE Vernova Hitachi Nuclear Energy

Wilmington, NC

#### Nuclear Fuel Performance and Design Intern

May 2023 - Aug 2023

- Constructed a Python-based comprehensive database for 200+ BWR fuel units to identify failure signatures across five reactor sites.
- Conducted root cause analysis (RCA) and failure assessments using thermal-mechanical models to identify high-risk reactor rod cells.
- Presented critical failure findings to thermal-mechanical team, enabling data-driven preventative inspections and NQA-1 compliance.

### FIRST Robotics (FRC) - Team 6517

Knoxville, TN

#### Robotics Team Lead

Aug 2016 - May 2020

- Directed design and manufacture of competitive robot components using SOLIDWORKS, Fusion 360, CNC milling, and 3D printing.
- Led the curriculum development for Computer Science and Robotics while mentoring 30+ students in robot design, C++, and Python.

## PUBLICATIONS

**Additive Manufacturing Letters** | Additive friction stir deposition of multi-layer aluminum-silicon carbide metal matrix composites

**Additive Manufacturing** | Structural stability of thin overhanging walls extrusion additive manufacturing of thermoset-based ink

**Additive Manufacturing** | Linking thermoset ink rheology to the stability of 3D-printed structures

**Journal of Manufacturing Processes** | A Dissimilarity Index for Monitoring Process in Additive Friction Stir Deposition

## TECHNICAL SKILLS

**Simulation & Analysis:** ANSYS Fluent, Mechanical, CFD, FEA, pyAnsys, topology optimization, heat transfer

**Programming & Controls:** Python (NumPy, OpenCV, TensorFlow), C++, MATLAB, Arduino, Closed-Loop Control

**Design & Manufacturing:** SOLIDWORKS (CSWA), GD&T, Additive Manufacturing (AFSD, FDM, SLA), CNC Milling

## AWARDS AND COMMUNITY SERVICE

**Academic:** Summa Cum Laude, 1794 Honors Scholar, Outstanding Senior & Junior in MAE, Order of the Engineer

**Service:** Keep Knoxville Beautiful, Habitat for Humanity